RNC

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name: RENEWCOILS ENGINEERING AND SUPPLY CO . LTD

WPS No.: RNC-WPS-014 Date: 29-Mar-13 Revision No.: 0

Supporting PQR No.(s): RNC-PQR-014 Date: 29-Mar-13 Type(s): Manual

Welding Process(es): Gas Tungsten Arc Welding + Shield Metal Arc welding (GTAW +SMAW)

JOINT (QW-402)

Joint Design: **Single V**

Backing (Yes) ____ (No) X

Backing Material (Type): N/A

☐ Metal ☐ Nonfusing Metal ☐ Nonmetallic ☐ Other

Notes: ____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. 5B Group No. 1 to P-No. 5B Group No. 1

Material Specification Type and Grade: A335 P5 , F182-F5, A234 WP5 TO A335 P5 , F182-F5, A234 WP5

Chem. Analysis and Mech. Prop.: ____

Thickness Range :

	Groove	Fillet
Base Metal :	1.6-18.54	ALL
Pipe Dia. Range :	2-7/8" and over	ALL

Other: ____

FILLER METALS (QW-404)	GTAW	SMAW
Spec. No. (SFA)	5.28	5.5
AWS No. (Class)	ER80S-B6	E8016-B6
F-No.	6	4
A-No.	4	4
Size of Filler Metals	Ø 2.4 ~ 3.2mm.	Ø 2.6 ~4.0 mm.
Deposited Weld Metals	3 mm.	6.27 mm.
Thickness Range :		
Groove	6 mm.	12.54 mm.
Fillet	ALL	ALL
Electrode-Flux (Class)	-	-
Flux Trade Name	-	-
Consumable Insert	-	-
Other	Kobelco - TGS-5CM	Kobelco - CM-5

POSITION (QW-405)

Position(s) of Groove: 6G

Welding Progression: Up X Down ____

Position(s) of Fillet: ALL

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range: 710°C~770°C ±10°C

Time Range: Holding Time 2 Hour

Notes: Heating Rate = 220°C Max/Cooling Rate = 250°C Max

PREHEAT (QW-406)

Preheat Temp. Min.: 200°C~250°C

Interpass Temp. Max.: 300°C

Preheat Maintenance: N/A

Notes: Resistance heater

GAS (QW-408)

	Gas(es)	(Mixture)	Flow Rate
	Percent Composition		
Shielding	Argon	9.99%	12-16 Ltr/Min
Trailing	-		-
Backing	Argon	9.99%	13-17 Ltr/Min

RNC

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-014 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current [X] AC [X] DC Polarity EN(GTAW)+ EP(SMAW)

Amps (Range) 70-90 (GTAW) + 90-145 (SMAW) Volts (Range) 8-13(GTAW) + 18-24(SMAW)

Tungsten Electrode Size and Type Ø 2.4 mm., Thoriated (EWTh-2)

(Pure Tungsten, 2% Thoriated, etc.)

Mode of Metal Transfer for GMAW N/A

(Spray arc, short circuiting arc, etc.)

Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS

Orifice or Gas Cup Size 8.0 mm - 19.0 mm

Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING

Method of Back Gouging N/A

Oscillation N/A

Contact Tube to Work Distance N/A

Multiple or Single Pass (per side) MULTIPASS

Multiple or Single Electrodes SINGLE ELECTRODE

Travel Speed (Range) 5-14 Cm/Minute

Peening N/A

Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER80S-B6	2.4 mm.	DC EN	80-120	8-12	5-7	
2nd	GTAW	ER80S-B6	2.4 mm.	DC EN	100-130	10-13	7-9	
3 & Over	SMAW	E8016-B6	3.2 mm.	AC/DCEP	80-130	18-24	9-14	

We,the undersigned, certify that the statements in this record are correct and that the test welds were prepared ,welded, and tested in accordance with the requirement of **ASME-IX, 2010 latest Revision.**

Prepared by : Mr.Chatchawan S. Name Signed 11/04/13 Date

Reviewed by : ภูชิน สีหอมไชย Phuchin Seehomchai Name Signed 11/04/13 Date

Approved by: ภูชิน สีหอมไชย Phuchin Seehomchai Name Signed 11/04/13 Date

Procedure Qualification Record (PQR)

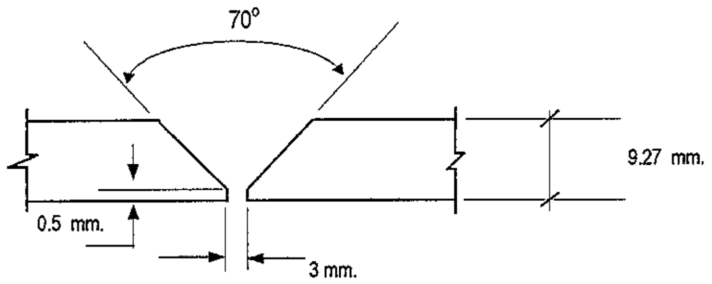
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-001

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-014	Date :	29 March 2013
WPS Number :	RNC-WPS-014		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic):	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)

Material Spec.	A335	To	A335
Type or Grade	P5	To	P5
P. No.	5B	To	5B
Thickness of Test Coupon	9.27 mm.		
Diameter of Test Coupon	ø 10"		
Others			

FILLER METALS (QW-404)

	GTAW	SMAW
SFA Specification	A 5.28	A 5.5
AWS Classification	ER80S-B6	E8016-B6
Filler Metal F-No.	6	4
Weld Metal Analysis A-No.	4	4
Size of Filler Metal	Ø 2.4 mm.	Ø 3.2 mm.
Others	Kobelco TGS-5CM	Kobelco CM-5
Weld Metal Thickness	-	

POSITION (QW-405)

Position of Groove	6G
Progression of Welding (Uphill or Down)	Up Hill
Others	N/A

PREHEAT (QW-406)

Preheat Temperature	220° C
Interpass Temperature (max.)	250° C
Others	N/A

POSTWELD HEAT TREATMENT (QW-407)

Temperature	720° C
Hold Time	2 HR
Others	Heating Rate 190° C/Hr
	Cooling Rate 240° C/Hr

GAS (QW-408)

	Percent Composition		
	Gas (es)	Mixture	Flow Rate
Shielding	Argon	99.95%	12 L/Min.
Trailing	-	-	-
Backing	-	-	-

ELECTRICAL CHARACTERISTICS (QW-409)

Current	DC (GTAW)	DC (SMAW)	
Polarity	EN	EP	
Amperage	122 - 129 (GTAW)	Voltage	10 (GTAW)
	100 - 121 (SMAW)		18 - 20 (SMAW)
Tungsten Electrode Size	ø 2.4 mm. 2% thoriated.EWTh-2		
Others	N/A		

TECHNIQUE (QW-410)

Travel Speed	5 - 9 cm/min.
String or Weave Bead	Both
Oscillation	N/A
Multiple or single Pass	Multiple Pass
Multiple or Single Electrodes	Multiple Electrodes
Others	N/A

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number-: RNC-PQR-014

Tensile Test Report No.TS-13-04-028

Specimen	Width (mm)	Thickness (mm)	Area (mm^2)	Ultimate Total Load (KN)	Ultimate Strength (N/mm^2)	Type of Failure & Loc.
1048/13 TR1	19.22	9.45	181.629	110.383	607.74	OUT OF WELD
1048/13 TR2	19.19	8.86	170.023	102.695	604.01	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-04-011

Type and Figure Number	Type of Bend	Indication	Results
1048/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
1048/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
1048/13 BR1	Transverse root bend	No Open Discontinuity	Accepted
1048/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170) Report No.IM-13-04-014

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Average (Joules)	Lateral Expansion Mils
1048/13-1	**Weld Metal**	**V-Notch**	**- 20 °C**	**81.57**		
1048/13-2	**Weld Metal**	**V-Notch**	**- 20 °C**	**74.23**	**81.87**	
1048/13-3	**Weld Metal**	**V-Notch**	**- 20 °C**	**89.81**		
1048/13-1	**HAZ**	**V-Notch**	**- 20 °C**	**261.76**		
1048/13-2	**HAZ**	**V-Notch**	**- 20 °C**	**286.52**	**267.97**	
1048/13-3	**HAZ**	**V-Notch**	**- 20 °C**	**255.62**		
1048/13-1	**Base Metal**	**V-Notch**	**- 20 °C**	**307.98**		
1048/13-2	**Base Metal**	**V-Notch**	**- 20 °C**	**311.5**	**307.86**	
1048/13-3	**Base Metal**	**V-Notch**	**- 20 °C**	**304.09**		

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A

Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR-001

Hardness Testing Report No. HV-13-04-005

Macrostructure Report No. MA-13-04-013

Welders Name Mr.Narongkon Y. Welder Number W-013

Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor : Inspected by : Mr.Wanlop N.

Date : 11/04/13 Certified by : Mr.Chaloemkiet R.

Date : 9 April 2013

QA/QC RNC ENGINEERING & SUPPLY CO.,LTD.

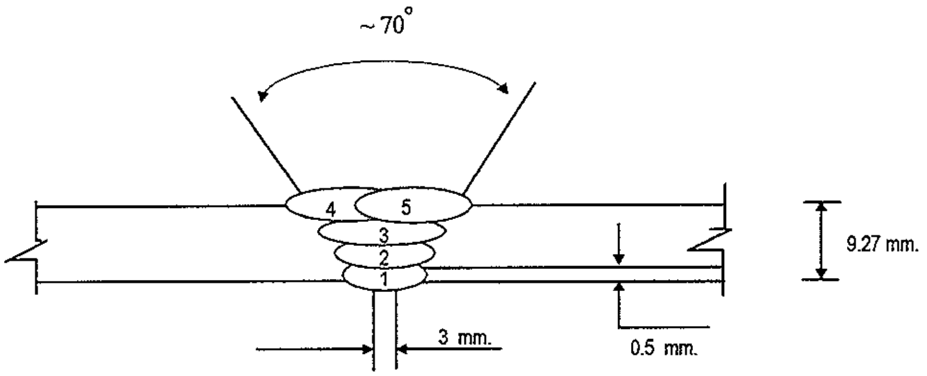
PAE TECHNICAL SERVICE

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-014
		WPS No. : RNC-WPS-014
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST
		CONDITION AND RESULT

Welding Sequences for RNC-PQR-014 (Butt Joint)



Pipe Dia. 10"

Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave Bead	Wire Feed Speed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
1	GTAW	ER80S-B6	2.4	122	10	DCEN	-	-	5	14.60	12	12
2	GTAW	ER80S-B6	2.4	129	10	DCEN	-	-	7	11.00	12	12
3	SMAW	E8016-B6	3.2	100	18	DCEP	-	-	9	12.00	-	-
4	SMAW	E8016-B6	3.2	121	20	DCEP	-	-	9	16.10	-	-
5	SMAW	E8016-B6	3.2	119	20	DCEP	-	-	9	16.10	-	-

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

PAE Technical Service Company Limited
Public Registration No. (5) 1571/2542
69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

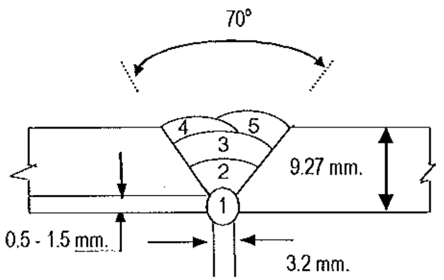
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 682208-9

WELDER PROCEDURE QUALIFICATION RECORD

Client : Renew Coils Engineering & Supply Co.,Ltd.	**Project Name :** MINI SHUTDOWN PTTGC	**Location :** RNC Work Shop	**Report No.** PQR-001
	Customer Order No. : N/A	**WPS No. :** RNC-WPS-014	**Date :** 29-Mar-13

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material Spec	Material Size	Electrode Specification	Results Visual	Results RT
W-013	MR.NARONGKON Y.	GTAW / SMAW	6G	ASME IX	A335 P5	ø 10" Thk. 9.27 mm.	ER80S-B6	ACCEPTED	ACCEPTED
							E8016-B6		

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER80S-B6	2.4	DCEN	122	10	5	220	14.6	Argon	12	Argon	12
2	GTAW	ER80S-B6	2.4	DCEN	129	10	7	219	11.0	Argon	12	Argon	12
3	SMAW	E8016-B6	3.2	DCEP	100	18	9	219	12.0	N/A	N/A	N/A	N/A
4	SMAW	E8016-B6	3.2	DCEP	121	20	9	220	16.1	N/A	N/A	N/A	N/A
5	SMAW	E8016-B6	3.2	DCEP	119	20	9	220	16.1	N/A	N/A	N/A	N/A

PAE Inspection By	Witness By	Approved By
SIGNATURE :	SIGNATURE :	SIGNATURE :
NAME : Mr.Wanlop N.	NAME : Mr. Chatchawan S.	NAME : ภูชิน สีหอมไชย
DATE : 30-Mar-2013	DATE : 11/04/13	DATE : Phuchin Seehomchai

QA/QC RNC

RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

PAE TECHNICAL SERVICE

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)682208-9, Fax : 038)682206, SRIRACHA : 038)330220, 330229

REPORT No.	PQR-001	***RADIOGRAPHIC EXAMINATION REPORT***	INSPECTION & TEST RECORDS
DWG.No.	-		
DATE :	1-Apr-13		

CLIENT :	Renew Coils Engineering & Supply Co.,Ltd.	PROJECT NAME :	MINI SHUTDOWN PTTGC

PROCEDURE No. :	PAE-RT-001 (Rev.0)	EXPOSURE TIME	7 min. 20 sec.	SOURCE TO FILM DISTANCE	10" Inch. / - mm.
RADIATION SOURCE	X-ray (-) MAX - kVG. / γ - ray (Ir-192)	DENSITY RANGE	2.0 - 3.5	TYPE OF PENETRAMETER	ASTM-1B
		SENSITIVITY	2%		
SOURCE RANGE	12.50 Ci.	FILM TYPE	KODAK "MX"	FILM PROCESSING AUTO	-
SOURCE SIZE	3 x 3 mm.	FILM SIZE	3.5" x 17"	FILM PROCESSING MANUAL	✓
TUBE VOLTAGE	- kVP.	INTENSIFYING SCREEN FRONT	0.005 Inch.	MATERIAL	A335 P5
TUBE CURRENT	- mA	INTENSIFYING SCREEN BACK	0.005 Inch.	THICKNESS	9.27 mm.
APPLICABLE STANDARD :	ASME Section V Article 2			ACCEPTANCE CRITERIA :	ASME-IX

RADIOGRAPHIC TECHNIQUE : ☐ SWSI ☑ DWSI ☐ DWDI

NAME / WQT	Position No.	Pipe/Plate Size Dia	Pipe/Plate Size Thk(mm)	Welder No.	Film No.	Interpretation	Results	Remarks
GTAW+SMAW / WPS No.RNC-WPS-014								
MR.NARONGKON Y.	6G	10"	9.27	W-013	A-B	NAD	ACC	After PWHT
					B-C	P ø 0.5 mm.	ACC	
					C-A	NAD	ACC	

Total 3.5"x8.5" - Films

3.5"x17" 3 Films

ABBREVIATIONS :

ACC : Accepted	WH : Worm Hole	LP : Lack of Penetration	
REJ : Rejected	C : Crack	LF : Lack Of Fusion	CRP : Concave Root Penetration
CP : Cluster Porosity	P : Porosity	IP : Incomplete Penetration	ERP : Excess Root Penetration
SF : Surface Flaw	UC : Undercutting	TI : Tungsten Inclusion	NAD : No Apparent Defects
EUC : External Udercut	SL : Slag Inclusion	BT : Burn Through	RS : Reshoot

PAE INSPECTED BY	REVIEWED BY	APPROVED BY
SIGNATURE :	SIGNATURE :	SIGNATURE :
NAME : Mr.Sarawut P.	NAME : Mr.Chatchai	NAME : ภูชิน สีหอมไชย Phuchin Seehomcha
DATE : 2-Apr-13	DATE : 1/04/13	DATE :

RNC
RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

WELDERS CERTIFICATE

Record of Welder or Welding Operator Qualification Test

ptt GLOBAL CHEMICAL

Client : PTT GLOBAL CHEMICAL CO., LTD. Project Name : PTTGC 2013

Contractor : Renewcoils Engineering & Supply Co.,Ltd.. Sub-Contractor : -

Welders Name : Mr. Narongkorn Y. I.D. No. : 1 2512 00008 13 4 Welder No. : W-013

Welding Process (es) used : GTAW/SMAW Type : Manual

WPS Number (Used by welder during the welding of test coupon) : RNC-WPS-014

Base material (s) welded : ASTM A 335 P5 Thickness : 9.27 mm. (PQR)

(QW - 350)	Actual Values	Range Qualified
Backing (metal, weld metal, welder from both sides, flux, etc.) (QW-402)	With out Backing	With / With out Backing
ASME P - Number to ASME P- Number (QW - 403)	P-5B + P-5B	P - No. 5B
Base metal Dimension ☐ Plate ■ Pipe (include diameter)	OD 10"	O.D 2-7/8" to Unlimited
Filler metal specification (SFA) A 5.28/5.5 Classification (QW - 404)	ER 80S-B6/E8016-B6	F no.6+4
Filler metal F - Number	F6+4	F6+4
Consumable insert for GTAW or PAW	N/A	N/A
Weld deposit thickness for each welding process : GTAW /SMAW	3 /6.27 mm.	Max. 6 / 12.54 mm
Welding position (QW - 405)	6G	All Position Unless Downhill
Welding Progression (uphill / downhill) (QW - 405)	Uphill	Uphill
Backing gas for GTAW , PAW , or GMAW ; Fuel gas for OFW (QW-408)	NO	YES or NO
GMAW transfer method / mode (QW - 409)	N / A	N / A
Welding current type and polarity	DC / EN	DC / EN

Visual Inspection Results (QW - 302.4)

Fit - up :	Root Pass :	Weld Profile :
Acceptable	Acceptable	Acceptable

Guided Bend Test Results

Guided Bend Test Type	( ) QW - 462.2 (Side) Results	( ) QW - 462.3a (Trans. R&F Type)	( ) QW - 462.3b (Long. R&F) Results
-	-	-	-

Radiographic Test Results (QW - 304 & 305) ACCEPTABLE SEE REPORT No.PQR-001

(For alternative qualification of groove welds by radiography)

Fillet Weld - Fracture Test Result - Length and percent of defects - in.

Macro test fusion - Fillet leg size - in.x - in. Concavity / Convexity - in.

Welding test conducted by PAE TECHNICAL SERVICE CO.,LTD.

Mechanical test conducted by N / A Laboratory test No. N / A

We certify that the statements in this record are correct and that the test coupons were prepared . Welded and tested in accordance with the requirement of : ASME - SECTION IX : 2010 / ASME B 31.3 : 2010

	Prepared By	Reviewed By / Approved By	Approved By
Signature :	(signature)		
Name :	Mr. Chatchawan-S		
Date :	11/04/13		

BOOK NO. 020 PAGE NO. 0967

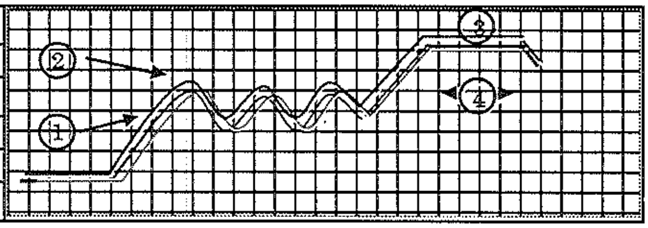
GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.

PREHEAT REPORT

CLIENT :	RENEW COILS ENGINEERING & SUPPLY CO.,LTD.
PROJECT :	MINI SHUT DOWN PTTGC
CONTRACTOR:	GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.
PREHEAT METHOD:	○ GAS ☑ ELECTRIC COMPLETION DATE : 29-03-2013
LOCATION:	☑ SHOP ○ FIELD REPORT NO. PH -001
RECORDER NO:	EH- 07ZA194 PAGE NO. 1/2
CHART SPEED:	☑ 25 MM/Hr ○ 50 MM/Hr ○ 100 MM/Hr

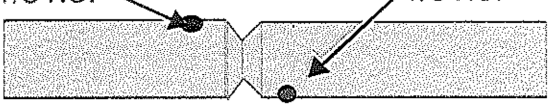
RESULT RECORDER CHART TO BE ATTACHED

DESCRIPTION	REQUEST	ACTUAL
1) PREHEAT TEMP (C° MIN)	200-250	220
2) INTERPASS TEMP (C° MAX)	300	250
3) POSTHEAT TEMP (C°)	-	-
4) POSTHEAT TIME (Hr)	-	-
MATERIAL : A335 P5		



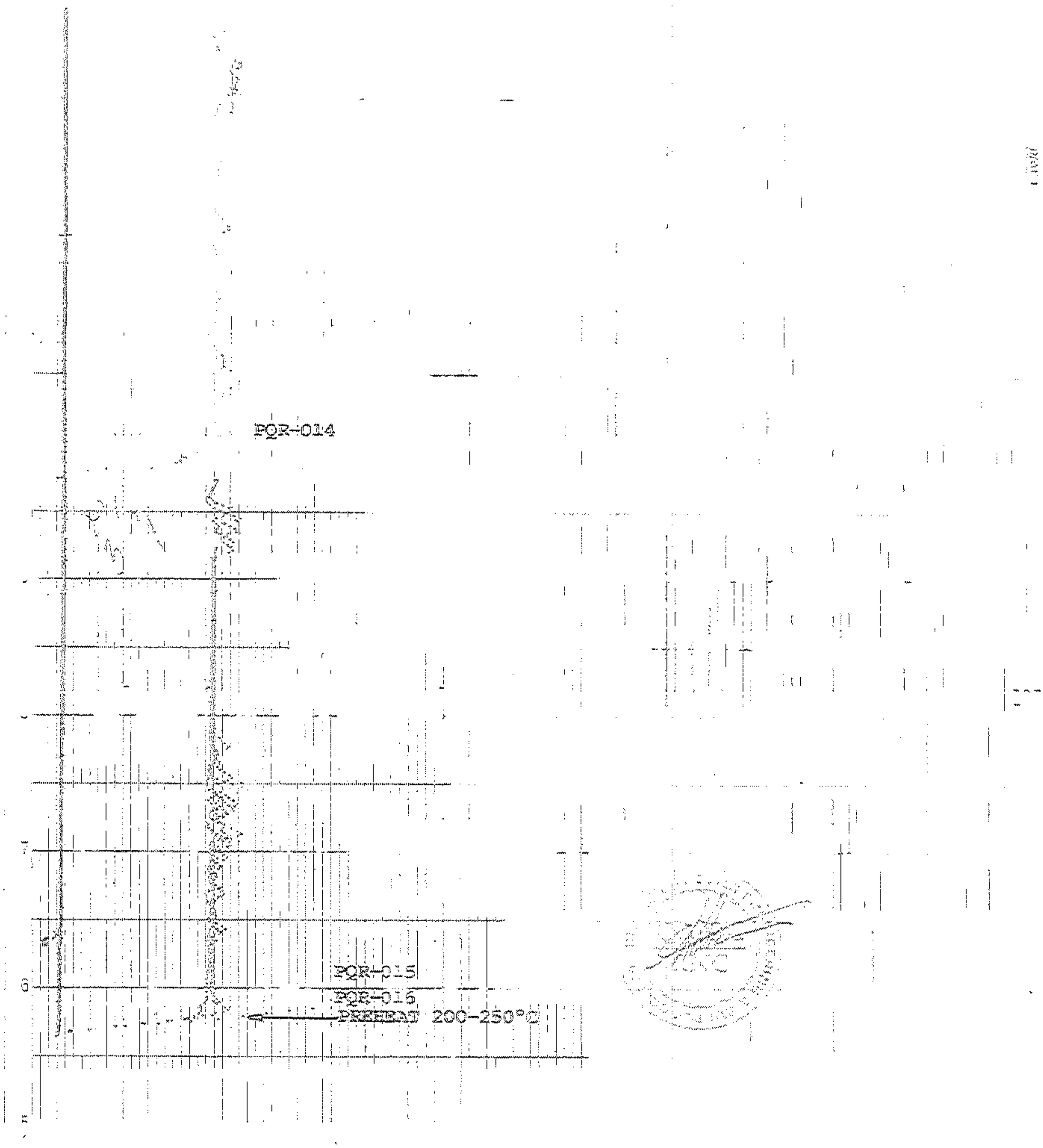
LINE NO.& CLASS & SHT NO.& SPL.NO. DWG.NO & TANK NO.	WELD NO.	SIZE	THK. (MM)	WELD TYPE	T/C NO.	REMARK
PQR NO. RNC-PQR-014		10"	9.27	BW	3,4	

T/C NO. T/C NO.



TOTAL	1	JOINTS.
TOTAL	10	DB.

CHECKED BY GLOBAL HEAT TREATMENT (THAILAND)	APPROVED BY	APPROVED BY
SIGN:	SIGN:	SIGN:
NAME: KOMKHUN D.	NAME: Mr. Chatchawan S.	NAME: ภูชิน สีหอมไชย Phuchin Seehomchai
DATE: 29-03-2013	DATE: 2/04/13	DATE:


PQR-014
PQR-015
PQR-016
PREHEAT 200-250°C

LPN
LPN PLATE MILL
Testing Laboratory LPN Plate Mill Public Company Limited
199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.
Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

8941

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-04-028

Revision No. 0

Test Product	WPS No. RNC-WPS-014
Project Name	MINI SHUTDOWN PTTGC
Project No.	N/A
Material Specification	A335 P5
Dimension	Ø 10" (9.27 mm Thk.)
Date Received	03 April 2013
Date Tested	05 April 2013

Material Characterization	Pipe
Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Standard of Test	ASME Sec. IX : 2010 edition
Type of Test	Reduced Section Tension
Equipment's Capacity	200 TONS
Equipment / Serial No.	SHIMADZU UH200A / 32098926
Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm^2)	Gauge Length (mm)	Load (kN)	Strength (N/mm^2)	Load (kN)	Strength (N/mm^2)		
1	1048/13 TR1	N/A	9.45	19.22	181.629	N/A	N/A	N/A	110.383	607.739	N/A	OUT OF WELD
2	1048/13 TR2	N/A	8.86	19.19	170.023	N/A	N/A	N/A	102.695	604.007	N/A	OUT OF WELD

The unit of N/mm^2 is equivalent to MPa

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : MR.NARONGKON Y.

Welder No. : W-013

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date ...06.../...04.../...13...

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date ...../...04.../...13.....

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC

LPN PLATE MILL

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

28944

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-04-011

Revision No. 0

Test Product	WPS No. RNC-WPS-014	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PTTGC	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P5	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 10" (9.27 mm Thk.)	Diameter of Meudrel	36 mm
Date Received	03 April 2013	Bend Angle	180 Degree.
Date Tested	05 April 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	1048/13 BF1	Transverse face bend	9.41x38.08x161.07 mm	No Open Discontinuity	N/A	-
2	1048/13 BF2	Transverse face bend	9.24x38.07x163.55 mm	No Open Discontinuity	N/A	-
3	1048/13 BR1	Transverse root bend	8.81x38.07x162.91 mm	No Open Discontinuity	N/A	-
4	1048/13 BR2	Transverse root bend	9.15x38.08x162.11 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : MR.NARONGKON Y.

Welder No. : W-013

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date ....9..../....4..../..13....

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date ....9..../....4..../..13....

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28947

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-04-014

Revision No. 0

Test Product	WPS No. RNC-WPS-014	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PTTGC	Test Method	ASTM A370-10 and ASTM E23M-07a[e1]
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P5	Type of Test	Charpy V-Notch Impact
Dimension	Ø 10" (9.27 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	03 April 2013	Equipment 's Capacity	542 Joules
Date Tested	05 April 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension Thickness (mm)	Width (mm)	Length (mm)	Results Energy Absorbed (Joules)	Average (Joules)	Location of Test	Remark
1	1048/13-1	-20°C	7.507	10.007	54.76	81.57	81.87	Weld Metal	-
2	1048/13-2	-20°C	7.507	10.007	54.79	74.23			
3	1048/13-3	-20°C	7.505	10.008	54.82	89.81			
4	1048/13-1	-20°C	7.505	10.003	54.77	261.76	267.97	HAZ	-
5	1048/13-2	-20°C	7.505	10.006	54.76	286.52			
6	1048/13-3	-20°C	7.497	10.007	54.74	255.62			
7	1048/13-1	-20°C	7.504	10.004	54.88	307.98	307.86	Base Metal	-
8	1048/13-2	-20°C	7.504	10.007	54.88	311.50			
9	1048/13-3	-20°C	7.504	10.007	54.87	304.09			

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : MR.NARONGKON Y.

Welder No. : W-013

Reported by : ....
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date ...01/04/13...

RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Approved by : ....
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date ...05/04/13...

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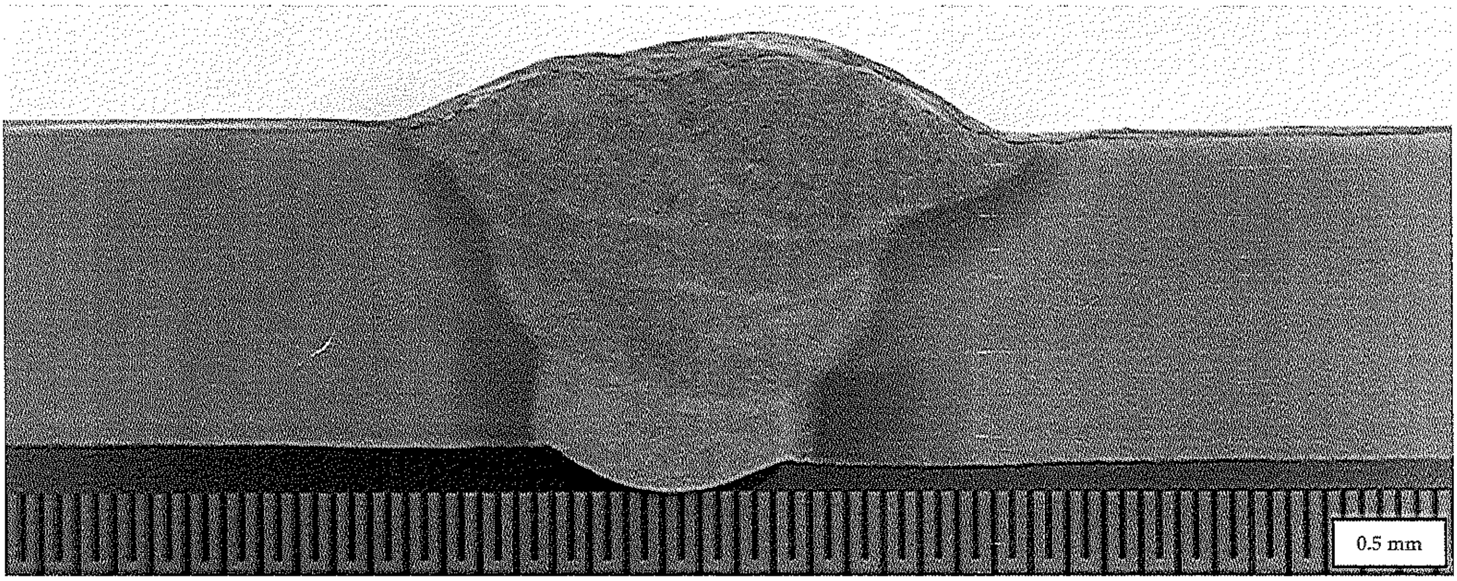
199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.
Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Macrostructure Analysis Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Report No. MA-13-04-013
Revision No.0

Test Product	WPS No. RNC-WPS-014	Material Characterization	Pipe
Test No.	1048/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A335 P5	Method of Preparation	ASME Secction IX : 2010 edition
Dimension	Ø 10" (9.27 mm Thk.)	Test Method	ASME Secction IX : 2010 edition
Welding Process/Position	GTAW+SMAW/6G	Standard of Test	ASME Secction IX : 2010 edition
Type of Joint	Butt Joint	Etchant Reagent	25 ml HNO_3 + 75 ml H_2O
Welder Name / No.	Mr. Narongkon Y. / W-013	Time of Etching	5 minutes
Date Received	3 April 2013	Date Tested	6 April 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : Project Name : MINI SHUTDOWN PTTGC

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD.
QA/QC
RNC

Reported by : ....
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 09 / 04 / 13

Approved by : ....
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 09 / 04 / 13

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

13291

Hardness Testing Report

Report No. HV-13-04-005
Revision No. 0

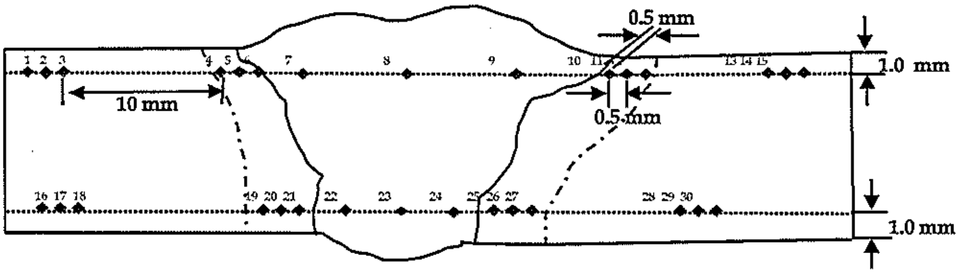
Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	WPS No. RNC-WPS-014
Test No.	1048/13 HN
Project Name	MINI SHUTDOWN PTTGC
Project No.	N/A
Material Specification	A335 P5
Dimension	ø 10" (9.27 mm Thk.)
Welding Process/Position	GTAW+SMAW/6G
Type of Joint	Butt Joint
Date Received	3 April 2013
Date Tested	8 April 2013

Material Characterization	Pipe
Test Method	ASTM E92-82 (Reapproved 2003)$^{\varepsilon 2}$
Standard of	ASME Section IX : 2010 edition
Type of test	Vicker Hardness Test
Equipment	Wilson Wolpert , 432 SVD (V32D219)
Indentor Size	Pyramid diamond, angle of 136°
Test Load	HV 10 Kgf.
Total Force Dwell Time	15 s
Cal. Block	No. A60023 No. A62072
Ambient Temperature	25.6 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)	Point No.	Location	Hardness value (Scale)
1	BASE	177.0	16	BASE	175.6		***************************	
2	BASE	175.5	17	BASE	177.4			
3	BASE	177.1	18	BASE	176.9			
4	HAZ	185.3	19	HAZ	230.8			
5	HAZ	217.8	20	HAZ	239.9			
6	HAZ	243.2	21	HAZ	245.8			
7	WELD	218.5	22	WELD	203.2			
8	WELD	229.3	23	WELD	205.0			
9	WELD	220.7	24	WELD	203.5			
10	HAZ	244.6	25	HAZ	240.1			
11	HAZ	227.4	26	HAZ	245.9			
12	HAZ	192.2	27	HAZ	242.1			
13	BASE	175.9	28	BASE	177.2			
14	BASE	177.8	29	BASE	178.7			
15	BASE	176.0	30	BASE	178.0			

Additional details : Welder Name: Mr. Narongkon Y. / W-013

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Reported by : ....
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 09 / 04 / 13

Approved by : ....
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 09 / 04 / 13

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BOOK NO. 118 PAGE NO. 5862

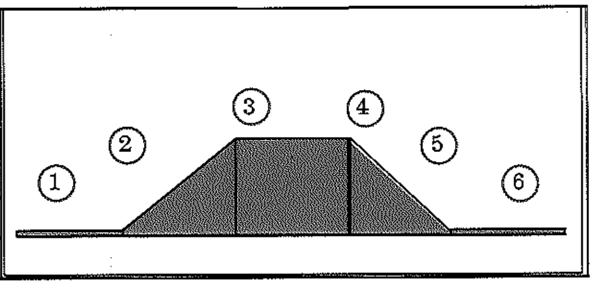
GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.

HEAT TREATMENT REPORT

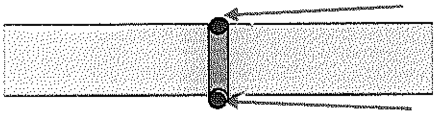
CLIENT :	RENEW COILS ENGINEERING & SUPPLY CO.,LTD.
PROJECT :	MINI SHUT DOWN PTTGC
CONTRACTOR:	GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.
DESCRIPTION:	☒ PWHT ○ NORMALIZE ○ ANNEALING ○ STABILIZE
LOCATION:	☒ SHOP ○ FIELD REPORT NO. PW-002
RECORDER NO:	EH- 07ZA194 PAGE NO. 1
CHART SPEED:	☒ 25 MM/Hr ○ 50 MM/Hr ○ 100 MM/Hr
METHOD BY:	☒ ELECTRIC ○ GAS ○ OIL ☒ LOCAL ○ FURNACE ○ INTERNAL
COMPLETION DATE:	30-03-2013 MATERIAL : A335 P5

HEAT TREATMENT CONDITION / RESULTS

DISCRIPTION	REQUEST	ACTUAL
1) LOADING TEMP (C°)	300	300
2) HEATING RATE (C°/Hr)	220	190
3) HOLDING TEMP (MIN-MAX)	710-770	720
4) HOLDING TIME (Hr)	2 HR.	2 HR.
5) COOLING RATE (C/Hr)	250	240
6) UNLOADING TEMP (C°)	300	300



LINE NO.& CLASS & SHT NO.& SPL.NO. DWG.NO & TANK NO.	WELD NO.	SIZE	THK. (MM)	WELD TYPE	T/C NO.
PQR NO. RNC-PQR-014		10"	9.27	BW	1,2

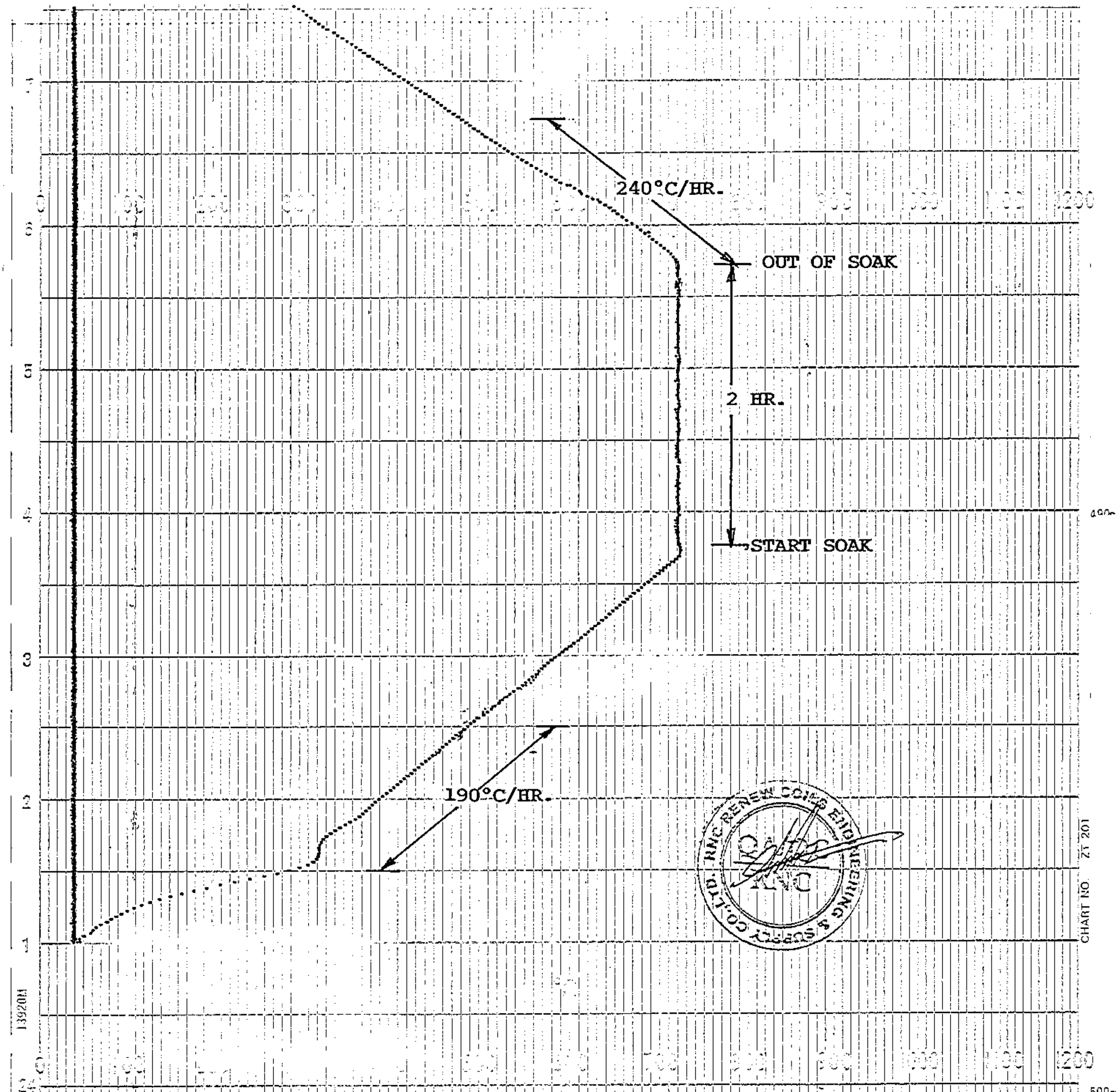


T/C NO.

T/C NO.

TOTAL	1	JOINTS.
TOTAL	10	DB.

CHECKED BY GLOBAL HEAT TREATMENT (THAILAND)	APPROVED BY	APPROVED BY
SIGN:	SIGN:	SIGN:
NAME: KONKHUN D.	NAME: Mr. Chatchawan S.	NAME: ภูชิน สีหอมไชย
DATE: 30-03-2013	DATE: 4/04/13	DATE: Phuchin Seehomchai



DATE. 30-03-2013

CHART NO. PW - 002

RECORDER NO. EH 07ZA194

CHART SPEED.25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

PROJECT. MINI SHUT DOW PTTGC

PQR NO. RNC-PQR-014

SIZE. 10" THK. 9.27 mm.

MATERIAL. A335 P5

START TIME. 17.00 PM.

TC	COLOUR
1.	RED
2.	BLACK

PAE TECHNICAL SERVICE COMPANY LIMITED
NDT, Testing, Inspection & Engineering Consultant
69 Soi On-nuch 64 (Suksamarn), Srinakarin Road, Suanluang, Bangkok. 10250 THAILAND
Tel : BKK : 02)322-0222, Rayong : 038)682208-9, Sriracha : 038) 768923, 768924

POSITIVE MATERIAL IDENTIFICATION REPORT

Client : Renew Coils Engineering & Supply Co.,Ltd.	**Analyzer Model :** NITON XL3t 800	**Test Date :** 10-Apr-13
Project : Mini Shut Down PTTGC	**Analyzer Serial No.:** 39210	**Report No.** PMI-001
Procedure No. : PAE-PMI-007 Rev.0	**Location :** SHOP	**Attached Report :** -
WPS No. RNC-WPS-014		

Item	Dwg No. / Line No.	Joint No.	Size	Thk.	Welder No.	Alloy ID.	Chemical Compositions as Found (%)					Judgment	
							Cr	Mn	Fe	Mo	Ni	Yes	No
1.1	RNC-PQR-014 (Base 1)	-	10"	9.27	W-013	A335 P5	4.74	0.38	94.22	0.51	0.00	✓	
1.2	RNC-PQR-014 (Weld)	-	10"	9.27	W-013	A335 P5	5.05	0.40	92.96	0.62	0.00	✓	
1.3	RNC-PQR-014 (Base 2)	-	10"	9.27	W-013	A335 P5	4.59	0.52	94.15	0.49	0.00	✓	

Total : 3 Points

NOTE :
- The Composition material Analysis was carried out on Surface of material
- The result of inspection accordance with ASTM table1 specification.

Mat'l Spec.	Chemical Range		
	Cr.	Mo.	Mn.
A335 P5	4.00 - 6.00	0.45 - 0.65	0.30 - 0.60

	PAE INSPECTED BY	REVIEWED / WITNESSED BY	APPROVAL BY
SIGNATURE :			
NAME :	Mr. Anusom M.	Mr. Chatchawan S.	ภูชิน สีหอมไชย Phuchin Seehomchai
DATE :	10-Apr-13	10/04/13	

RENEW COILS ENGINEERING & SUPPLY RNC QC

19484/5

JC-04 1/ 1

SB99 JC96 F402913

INSPECTION CERTIFICATE

JFE

JFE Steel Corporation
CHITA WORKS
1,Kawasaki-cho 1-chome,Handa-City,Aichi Pref.475-8611
Japan
Date : AUGUST 30, 2012

Certificate No.(P/L No.) :PG1208Y613(75-6706)

Order No. :2J000KN-002

Shipper :MITSUI & CO., LTD.

Customer :

Reference No.:

Customer's Control No.1:

Customer's Control No.3 :KTC-12-99952

Commodity :SEAMLESS ALLOY STEEL PIPE NDE

P5

Project Name :

Effective Year:11/2011A

Cr 5.00

Specification:ASTM A 335 GRADE P5/ASME SA-335 GRADE P5(BVL)

Mo 0.51

Mn 0.39

10/2/13

Size :NPS10" X STD X 6000

Quantity : 15 Mass : 5,430 Kg Total Length :

Ship No.:

Construction No:

Chemical Composition (%) — H: Heat Analysis P: Product Analysis

MFG. No.	Heat No.		C	Si	Mn	P	S	Cu	Ni	Cr	Mo	Al	Ti	V	Nb	B	N	Sn		Ceq
			×1000	×100	×100	×1000	×1000	×100	×100	×100	×100	×1000	×1000	×1000	×1000	×10000	×10000	X1000		×100
		max H	150	50	60	25	25			600	65									
		min H			30					400	45									
		max P	150	50	60	25	25			600	65									
		min P			30					400	45									
60526-01 (15)	1-27485	H	110	32	39	15	3			464	49									
		P	110	34	39	15	2			464	50							<2		
		P	120	34	39	14	2			469	50							<2		

Hydrostatic Test	1200PSI GOOD	Bend Test	
		Flattening Test	GOOD
N.D.E.	UT:GOOD	Flaring Test	
Visual&Dimension	GOOD	Drift Test	
Tensile Specimen	Strip	ASTM 1-1/2"	
	Round		
	Full Sec.		

Tensile Test: Parent (A370) Weld ( ) Direction (L)GL(2")
Impact Test: Direction ( )
Hardness Test: Spec. HV

MFG. No.	Heat No.		Y.P. or Y.S. (KSI)	T.S. (KSI)	EL. (%)	Y.R.	T.S. Weld	I.V. or A.E. EACH	I.V. or A.E. AVE.	Shear Area(%) EACH	Shear Area(%) AVE.	Hardness HV
		max.										248
		min.	30.0	60.0	30							
60626-01 (16)	1-27485		63.1	83.2	37							190
			63.1	83.0	38							

(Remarks)
UT=A/SA335/E213:U-SHAPED NOTCH
HEAT TREATMENT CONDITION
NORMALIZING : 920°C X 20MIN.
TEMPERING : 790°C X 60MIN.
EN10204 3.1
NACE MR0175(HARDNESS ONLY)
PMI:GOOD

PTT
Chayut S.
Inspection QA/QC Division
Integrity and Reliability
15/2/2013

QA/QC RNC ENGINEERING & SUPPLY CO.,LTD.

WE HEREBY CERTIFY THAT THE MATERIALS DESCRIBED HEREIN HAVE BEEN MANUFACTURED, INSPECTED AND TESTED IN ACCORDANCE WITH THE CUSTOMER'S SPECIFICATION(S), AND THAT THEY SATISFY THE REQUIREMENTS.

唐鎌勝則
Manager,Quality Assurance Sec.

PC